

AdaBoost algorithm

Evert Jan Karman

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1 The algorithm

0. We have N data points, assign each data point the same weight $\alpha_i = 1/N$

We have T weak learners (decision stumps)

For $t = 0, 1, \dots, T-1$

1. Learn weak learner f_t
2. Compute weighted error ε_t
3. Compute weight w_t
4. Recompute α_i for $i = 0 \dots N - 1$
- 4a. Normalize α so that

$$\sum_{i=0}^{N-1} \alpha_i = 1$$

2 Formula for 2. weighted error

$$\varepsilon_t = \sum_{i=0}^{N-1} \alpha_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$$

3 Formula for 3. weight

$$w_t = \frac{1}{2} \ln \frac{1 - \varepsilon_t(f_t)}{\varepsilon_t(f_t)}$$

Intuition: w_t measures the importance of the current weak learner. A lower weighted error gives it a larger weight.

4 Formula for 4. new α

$$\alpha_i^{\text{new}} = \begin{cases} \alpha_i \cdot e^{-w_t} & \text{if } \hat{y}_i = y_i \\ \alpha_i \cdot e^{w_t} & \text{if } \hat{y}_i \neq y_i \end{cases}$$

Intuition: This is increasing the weight for data points that have been misclassified by current learner.